Discussion Report on the Tangent Gauss Map

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The starting point of our discussion was the following question. Let A be a simple abelian variety of dimension $n > 2$, and let $C \subset A$ be an irreducible curve which generates A. Is the tangent Gauss map

$$\gamma_C \colon C^{\mathrm{sm}} \longrightarrow \mathbb{P}^{n-1}$$

generically injective unless the failure of injectivity is explained by a nontrivial symmetry of C induced by an affine automorphism of A whose linear part acts by a scalar?

Our recent progress shows that this expectation is false. We construct a simple abelian threefold A and a generating curve $C \subset A$ for which the tangent Gauss map has generic degree 3, while no nontrivial affine automorphism of A with scalar differential preserves C.

We also discussed the main idea behind the construction. It starts from a special genus 4 hyperelliptic curve arising from a tangential cover of an elliptic curve. We then deform this example to a non-hyperelliptic curve while preserving the Hodge-theoretic structure needed to obtain the corresponding abelian threefold. This leads to the desired counterexample and shows that non-injectivity of the tangent Gauss map can arise from the intrinsic geometry of the curve, rather than from symmetries of the ambient abelian variety.